

Function Matrix

Beamer Slides DSL Template

Hypo-LaTeX

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Function Matrix Overview

Slide Goal

Connect a function's domain, codomain, kernel, image, and rank to the matrix used to represent a linear map.

- A function matrix records how input coordinates determine output coordinates.
- The public template is synthetic and contains no private course or corpus text.

Function Matrix Overview (continued 2/3)

- Separator-driven frames inherit the previous H3 frame title while the inheritance limit allows it.
- Use this pattern only when a topic needs one short continuation frame.

Domain And Codomain

Notation

For a linear map $T : V \rightarrow W$, the domain is V and the codomain is W .

- Matrix columns correspond to basis vectors from the domain.
- Matrix rows correspond to output coordinates in the codomain.

Kernel, Image, Rank

Reader Check

Use the matrix to locate the kernel, image, and rank before drawing conclusions about injectivity or surjectivity.

- Kernel: all input vectors mapped to zero.
- Image: all reachable output vectors.
- Rank: the dimension of the image.

Function Matrix Table

表: Function matrix evidence

Object	Matrix evidence	Slide check
Domain	Number of input coordinates	Match column count
Codomain	Number of output coordinates	Match row count
Kernel	Solutions of $Ax = 0$	Detect null vectors
Image	Span of columns of A	Count pivot columns
Rank	Dimension of image	Compare with target dimension

Local Asset Rule

Asset Discipline

Slide decks may reference local assets with relative paths, but public templates must not fetch remote or private files during builds.

- Keep public examples self-contained unless a local fixture asset is committed.
- Use `resource-root` only for local project assets that belong with the deck.
- Image dimensions preserve aspect ratio by default; add `stretch=true` only when an image should deliberately fill both width and height.
- Standalone caption-less image lines are centered automatically with a `0.75\textheight` height cap; place side-by-side figures on one line and give each a `width=NN%` that sums below 100%.